
Porting SAMI to Audio: Unsupervised Pitch–Timbre Disentanglement of Musical Notes via Score-Based Guidance

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Abstract

The Score-based Autoencoder for Multiscale Inference (SAMI) (Lyo et al., 2025) replaces a VAE’s deterministic decoder with a *frozen* diffusion model: the encoder steers denoising through a guidance gradient, so disentanglement emerges from the score mechanism, not a reconstruction bottleneck. We port it from images to audio, for the unsupervised disentanglement of *pitch* and *timbre* in NSynth dataset (Engel et al., 2017) notes. On a held-out, instrument-disjoint split the latent linearly encodes pitch ($R^2 = 0.88$) and timbre (kNN accuracy 0.97), separating the two factors better than a β -VAE on timbre, MIG family and orthogonality. A demo preserves timbre and shifts pitch in the right direction, though generative pitch-control amplitude stays limited.

1. Introduction & Related work

Learning factorized audio representations without supervision is an open problem. For an instrumental note the natural factors are pitch and timbre: pitch is a multiscale property, since the fundamental and its harmonics span the whole spectrum, and timbre is the spectral envelope that identifies the instrument

VAEs (Kingma & Welling, 2014) and β -VAEs (Higgins et al., 2017) learn such representations but suffer from **posterior collapse** (expressive decoders ignore the latent, $\mu \rightarrow 0, \sigma^2 \rightarrow 1$). SAMI replaces the decoder with a frozen diffusion model (Sohl-Dickstein et al., 2015; Ho et al., 2020), though, as we show, KL-side instability remains and requires control. Unlike LSGM (Vahdat et al., 2021), which diffuses within the latent space, we apply diffusion directly in the data space.

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The decoder is the denoiser: reconstruction uses guided reverse diffusion. The latent z influences generation via a guidance gradient, akin to classifier guidance (Dhariwal & Nichol, 2021). However, the conditioning signal is not a fixed label but the encoder’s output z ; the gradient flows back into the encoder, turning sampling into a learning signal.

We present the first audio port of SAMI. Using NSynth mel-spectrograms, we demonstrate unsupervised linear separation and manipulation of pitch and timbre, developing the diagnostic methodology that made this possible .

2. Method

Data. We use the four NSynth families with distinct, well-sampled acoustic timbres (guitar, keyboard, string, brass), restricted to MIDI pitch [48, 84] (C3–C6, 61,531 notes). This range is the register where all four families have real sampled notes and the fundamental stays within the mel band ($f_{\max} = 8$ kHz), so CREPE (Kim et al., 2018) can estimate it reliably. By restricting the data to an overlapping pitch register, we prevent the model from using pitch as a confounding shortcut to identify instruments. Timbre is then a clean 4-way factor with a 25% chance baseline. For evaluation we use the official NSynth *test* split, filtered identically (966 notes) and normalized with the *same* global constants as training (normalization is part of the model and is not recomputed).

Architecture. Inputs are log-mel spectrograms globally normalized to $[-1, 1]$. A mel encoder q_ϕ maps a log-mel spectrogram $x_0 \in \mathbb{R}^{1 \times 128 \times 256}$ to a Gaussian posterior $q_\phi(z | x_0) = \mathcal{N}(\mu_0, \sigma_0^2)$ with $z \in \mathbb{R}^D$ ($D = 32$). A U-Net ϵ_θ is the unconditional denoiser. There is *no* separate $z \rightarrow x$ network: the “decoder” is ϵ_θ steered by the encoder through guidance.

Diffusion. With a cosine noise schedule (Nichol & Dhariwal, 2021) ($\bar{\alpha}_t, T = 1000$; chosen for a stable signal-to-noise ratio across t), the forward process is $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \epsilon \sim \mathcal{N}(0, I)$, and $\gamma_t = \sqrt{1 - \bar{\alpha}_t}$.

SAMI loss. We encode the clean input to $z = \mu_0 + \sigma_0 \odot \epsilon_z$ (reparameterization, to allow the calculation of the gradient) and the noisy x_t to (μ_t, σ_t^2) . The guidance gradient is

$$g_t = \nabla_{x_t} \log q_\phi(z | x_t),$$

$$\log q_\phi = -\frac{1}{2} \sum_d \left[\frac{(z_d - \mu_{t,d})^2}{\sigma_{t,d}^2} + \log \sigma_{t,d}^2 \right], \quad (1)$$

The denoiser is deviated additively and matched against the true noise:

$$\hat{\epsilon}(x_t, t, z) = \epsilon_\theta(x_t, t) - \gamma_t g_t, \quad (2)$$

$$\mathcal{L} = \underbrace{\mathbb{E}[\|\epsilon - \hat{\epsilon}\|^2]}_{\mathcal{L}_x} + \beta \underbrace{\sum_d \max(\text{KL}_d, \lambda)}_{\mathcal{L}_z}, \quad (3)$$

with $\text{KL}_d = \frac{1}{2}(\mu_{0,d}^2 + \sigma_{0,d}^2 - \log \sigma_{0,d}^2 - 1)$ and λ the per-dimension *free bits* (Kingma et al., 2016): below λ nat, a dimension pays no KL, which prevents the informative dimensions from being crushed to the prior. Since with free bits the post-floor $\mathcal{L}_z \approx 0$ even for a healthy encoder, the collapse guard monitors the *raw* KL ($\lambda = 0$).

Training regime. Training proceeds in two stages. First we pre-train the denoiser ϵ_θ alone, as an ordinary unconditional DDPM, until its samples are qualitatively sound (Fig. 1, App.). We then *freeze* its parameters and train the encoder within the full SAMI system. Because the guidance has leverage only at high noise (Fig. 2, App.), we bias the timestep distribution toward large t , sampling $t = \lfloor T \cdot \text{Beta}(4, 1) \rfloor$ instead of uniformly. Final hyperparameters (selected by the diagnostics below): $D = 32$, $\lambda = 0.5$, $\beta = 10^{-5}$, batch 16, Adam (3×10^{-5}).

Guided sampling. We generate with guided DDIM (Song et al., 2021), a deterministic sampler that needs only 50–200 steps instead of the full $T = 1000$; this speed matters because every step recomputes the guidance gradient g_t by autodiff. The latent conditions generation through a *guidance scale* s , the sampling-time analogue of classifier-free guidance (Ho & Salimans, 2021), which amplifies how strongly g_t steers the denoiser:

$$\hat{\epsilon} = \epsilon_\theta(x_t, t) - s \gamma_t g_t. \quad (4)$$

Measuring the latent. We assess the *unsupervised* code with *linear probes* (Alain & Bengio, 2017): a linear probe is too weak to recover a factor unless it is already linearly present in μ , so a high score credits the latent, not the probe. For pitch we fit an ℓ_2 -regularized regression (ridge (Hoerl & Kennard, 1970)) and report held-out R^2 . For timbre, a categorical 4-family factor, we use a k -nearest-neighbour classifier (Cover & Hart, 1967).

We also summarize each factor by its latent *direction* ($\hat{w}_p$ for pitch, $\hat{w}_f$ for timbre) the normalized difference between the mean latents of the factor’s top and bottom terciles (for family, of two reference instrument classes); their separability is $\cos(\hat{w}_p, \hat{w}_f)$, read against an empirical chance level: random unit vectors in $\mathbb{R}^D$ give $|\cos| \approx 0.14 \pm 0.10$ at $D = 32$, so only values well above this indicate shared structure.

Furthermore, to evaluate whether $\hat{w}_p$ actively controls generation, we report $\text{corr}(k, f_0)$, the correlation between a step k taken along $\hat{w}_p$ in the latent space and the resulting fundamental frequency f_0 of the generated audio. The traversal starts from the latent μ of a single held-out guitar note (MIDI 60) per model, standardized with the held-out reference statistics, and moves along $\hat{w}_p$ at seven unit steps $k \in \{-3, \dots, 3\}$ in standardized-latent space, generating with guided DDIM at scale $s = 1$ from the same initial noise; a step counts as pitched when the CREPE confidence exceeds 0.5. We also report MIG (Chen et al., 2018) (Mutual Information Gap), which measures *single-axis concentration* rather than overall linear decodability. To quantify our code’s distributed nature, explaining why probe and MIG scores inherently diverge, we report the participation ratio (PR), the effective number of active latent dimensions computed from the μ covariance eigenvalues.

We score SAMI and a β -VAE with the *same* battery of metrics (Table 1) for a standardized evaluation. Generated pitch is measured with CREPE for three purposes: to identify the latent pitch direction, to extract f_0 for the generative control correlation, and to track the pitch of the transferred notes. Griffin–Lim (Griffin & Lim, 1984) is used solely to invert the generated mel-spectrograms back into waveforms for qualitative listening.

3. Experimental results

Denoiser, then encoder. We first verify the frozen generator: unconditional DDPM samples show clear harmonic stacks and diverse pitches (Fig. 1, App.), confirming it is not the encoder’s bottleneck. Training the encoder on top of it, however, initially failed: within one epoch, the posterior collapsed ($\mu \rightarrow 0, \sigma^2 \rightarrow 1$) for any $\beta > 0$. The cause lies in the KL term $\mathcal{L}_z = \sum_d \text{KL}_d$, which faces two failure modes: at $\beta = 0$, the unconstrained posterior massively over-encodes; at any $\beta > 0$, the regularizer overpowers the guidance gradient and empties the latent. A standard warm-up ($\beta = 0$, then ramped) only worsens the subsequent collapse (Fig. 3, left).

Per-dimension free bits (Kingma et al., 2016) fix this: we grant each latent dimension a free information budget λ , replacing KL_d with $\max(\text{KL}_d, \lambda)$. Below λ nat, the KL gradient is zero, giving the guidance time to shape informative

dimensions before the regularizer acts, while unused ones still collapse to the prior. With $\lambda = 0.5$, the raw KL stabilizes at ~ 40 nat (Fig. 3, right).

The guidance acts only at high noise. To locate *where* the latent affects generation, we use the score-matching loss of Eq. (3) as a diagnostic probe on a *fixed*, freshly initialized encoder: at each noise level t we evaluate it with guidance ($\mathcal{L}_x^g$, i.e. $\hat{\epsilon} = \epsilon_\theta - \gamma_t g_t$ as in training) and without it ($\mathcal{L}_x^u$, i.e. $\hat{\epsilon} = \epsilon_\theta$, the bare denoiser), and take their relative change $\Delta(t) = (\mathcal{L}_x^g - \mathcal{L}_x^u) / \mathcal{L}_x^u$ (Fig. 2). Δ grows from 0.4% at low noise to 408% at high noise: at low t the input is nearly clean and the denoiser ignores z , whereas at high t the latent strongly affects the prediction. At the freshly initialized encoder used for this diagnostic, the learning signal thus lives at high t ; since uniform sampling would spend most gradient steps at low noise, where the encoder receives none, we bias t toward high noise, $t \sim \text{Beta}(4, 1)$ (mean ≈ 0.8), a non-uniform weighting we adopted empirically, in the spirit of importance-sampled diffusion losses (Nichol & Dhariwal, 2021).

Final encoder. The encoder trains stably at $\beta = 10^{-5}$: the pitch probe margin is flat from epoch ~ 20 , and training is halted at epoch 31 by early stopping on the probe margin, calculated on a validation slice held out from training. Scored with the same battery as a β -VAE baseline trained at $D = 32$ on the identical data (Table 1), on the held-out test split SAMI wins on timbre (kNN 0.97 vs. 0.81), MIG family (0.10 vs. 0.04) and orthogonality of the pitch/timbre directions ($\cos \approx 0$ vs. 0.28).

The β -VAE leads on the raw pitch probe (R^2 0.91 vs. 0.88) and on MIG pitch (0.027 vs. 0.006): it concentrates pitch on fewer axes, but along a direction more entangled with timbre ($\cos$ 0.28 vs. 0.03). SAMI instead spreads pitch across axes (MIG 0.006, PR 0.30) while keeping it linearly decodable (R^2 0.88) and near-orthogonal to timbre.

Both codes are distributed (PR $\approx 0.3D$), hence low MIG overall; SAMI is nonetheless the more structured code, with markedly better timbre separation and near-orthogonal factor directions. The test split contains only 26 instruments (vs. 379 in training), giving more compact timbre classes, so the high absolute kNN reflects this easier setting as well as genuine generalization to unseen instruments. As for the pitch traversal experiment both models are direction-correct (corr 0.87 for SAMI, 0.89 for the β -VAE, over 7/7 pitched steps), but neither shows fine control on this single traversal: SAMI’s f_0 moved between two levels about four semitones apart, the β -VAE by under 1.5; from one traversal we cannot tell whether SAMI’s stepwise pattern is systematic or seed-dependent.

High correlation thus reflects monotonic direction, not us-

able control amplitude, consistent with the frozen-regime limitation.

Table 1. Latent representation quality, SAMI vs. β -VAE baseline ($D = 32$), on the official NSynth test split.

Metric	SAMI	β -VAE	Chance
probe R^2 (pitch) $\uparrow$	0.88	0.91	≈ 0
kNN accuracy (timbre) $\uparrow$	0.97	0.81	0.25
MIG (pitch) $\uparrow$	0.006	0.027	—
MIG (family) $\uparrow$	0.10	0.04	—
$ \cos(\hat{w}_p, \hat{w}_f) \downarrow$	0.03	0.28	0.14 ± 0.10
PR ($/D$)	0.30	0.34	—
corr(k, f_0) $\uparrow$	0.87 (7/7)	0.89 (7/7)	—

Timbre transfer. The transfer moves the latent along the *causal* pitch direction $\hat{w}_p$ (Sec. 2): with A (guitar) donating timbre and B (brass) pitch, we replace A ’s pitch component with B ’s, $z_{\text{new}} = \mu_A - \text{proj}(\mu_A, \hat{w}_p) + \alpha \text{proj}(\mu_B, \hat{w}_p)$, manipulated in standardized latent space, de-standardized, and decoded from the same initial noise x_T as A and B (paired comparison). As a qualitative check, we run the transfer on training-set notes, sweeping s and α and tracking the output pitch with CREPE. Timbre is preserved at low α (kNN majority on the training pool, which includes the source notes) but degrades as α increases; the best setting ($s = 5$, $\alpha = 0.3$) shifts pitch cleanly on 3/7 of the tested initializations (Fig. 4).

This success rate reveals a structural limitation rather than a tuning issue: with fixed input notes, changing only the seed (e.g., 1001 vs. 1004) flips the outcome, no matter how we tune s or α (increasing $s \geq 10$ simply degrades audio quality).

4. Discussion and Conclusion

In audio, SAMI learns an unsupervised latent space where pitch and timbre are linearly separable and generatively manipulable. On Table 1 SAMI yields a more structured latent than the β -VAE (timbre, MIG family, near-orthogonal directions): the β -VAE stores more raw pitch information, but along a timbre-entangled direction. Generative pitch control is direction-correct for both, stepwise and of limited amplitude (Sec. 3).

Its limit is generative: the frozen denoiser cannot always overcome the initial noise x_T , so on training-set notes the transfer is directionally correct but seed-dependent, clean on 3 of 7 initializations.

Future work will explore joint (non-frozen) training to remove this seed dependence, plus a held-out transfer and stronger timbre metrics.

5. Statement on the use of AI

AI tools, as large language models used through an interactive coding assistant, were used in this project as an implementation aid, always under the authors’ supervision. Their main role was assisting with code: drafting and debugging training and evaluation scripts (including the D=32 β -VAE baseline and its SLURM launcher on cluster, the paired metric-comparison scripts). They were also used for routine data processing and for minor editorial feedback on parts of the manuscript. The AI did not make modeling or methodological decisions, with two exceptions it explicitly suggested: the use of free bits to address the posterior collapse, and the non-uniform sampling of the timesteps (importance-weighted diffusion loss) during training. Apart from these two suggestions, model architecture and hyperparameters, the design of the diagnostic experiments, the choice and validation of evaluation metrics, and the interpretation of every result were decided by the authors on the basis of the experimental evidence. All suggestions were independently evaluated by the authors through controlled experiments before being adopted. All experiments were carried out by the authors on the HPC cluster of the computer science department, and all code and results were reviewed and verified by the authors, who take full responsibility for the submitted work.

Code. The source code for the project is available at https://github.com/Alegau03/SAMI_Audio

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Appendix

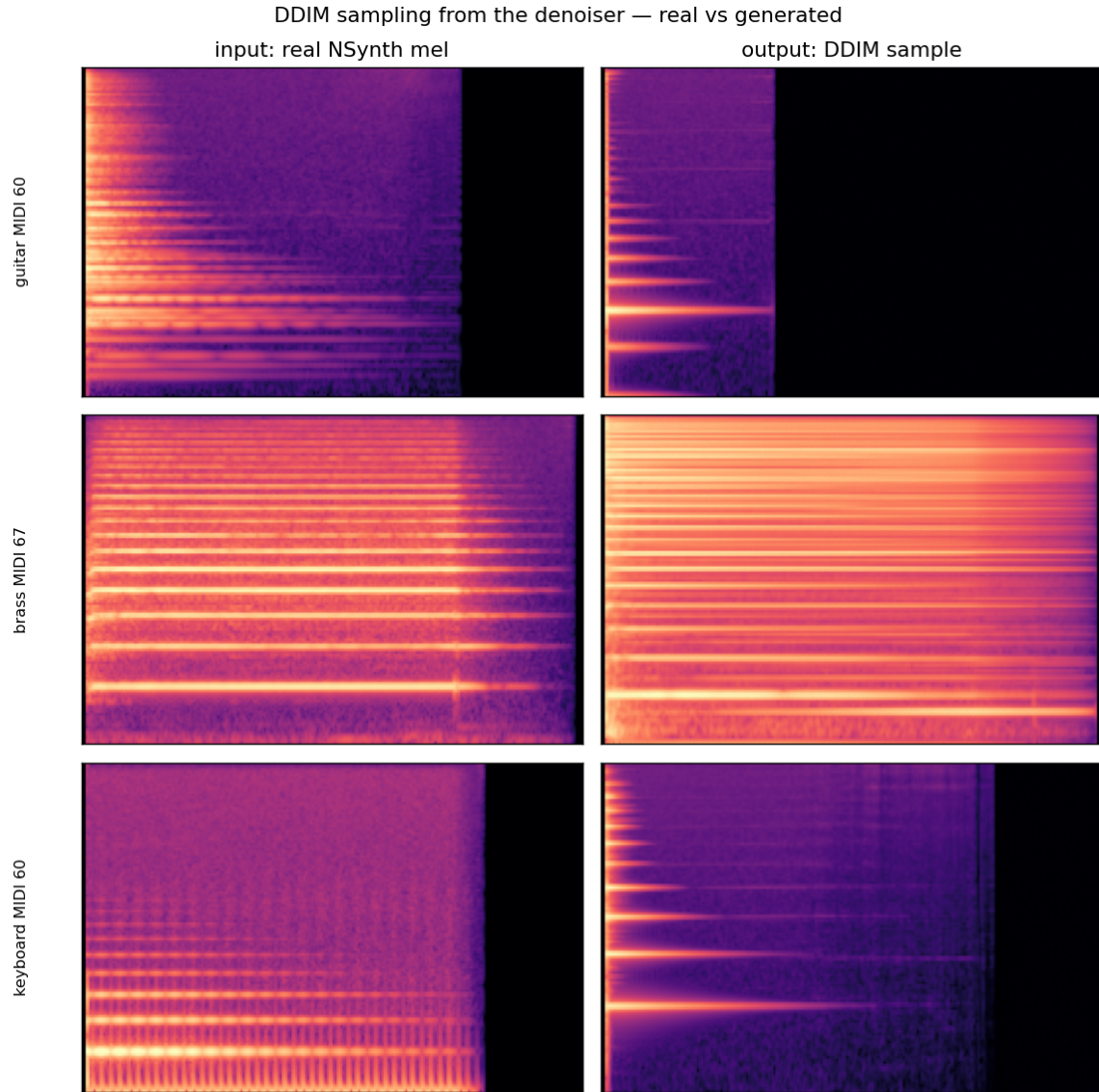


Figure 1. Unconditional denoiser samples. Unconditional DDIM samples from the frozen denoiser: harmonic stacks (fundamental + overtones), realistic temporal envelopes, and diverse pitches confirm the generator captures audio structure and is not the bottleneck for the encoder.

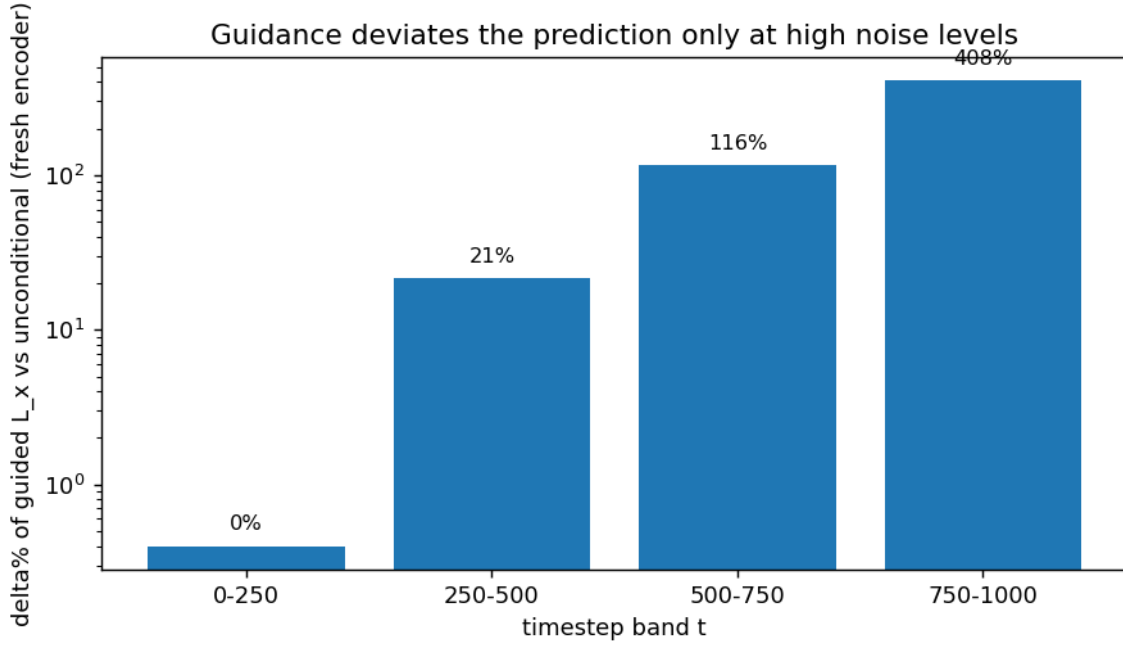


Figure 2. **Guidance leverage by noise level.** Relative difference $(\mathcal{L}_x^{\text{guided}} - \mathcal{L}_x^{\text{uncond}}) / \mathcal{L}_x^{\text{uncond}}$ per timestep quartile, on a fresh encoder: the guidance deviates the prediction only at high t , motivating the Beta(4, 1) timestep oversampling.

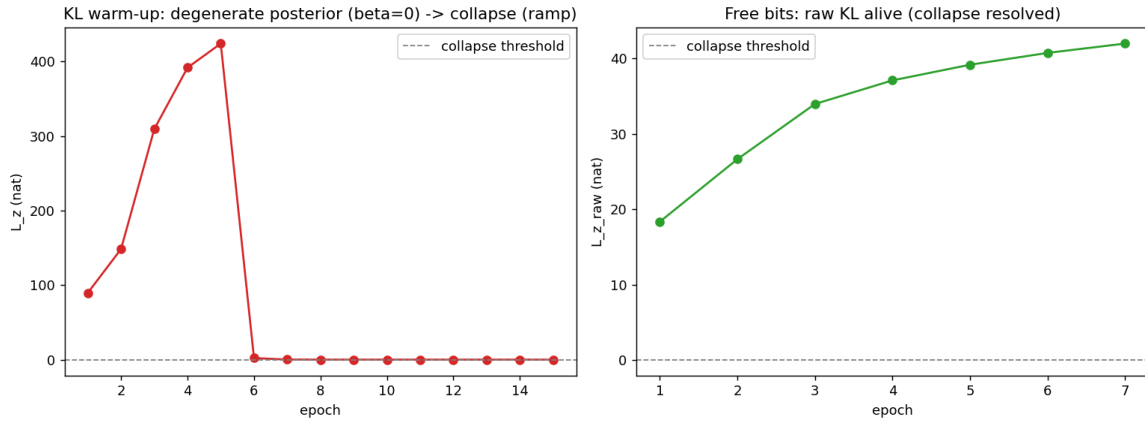


Figure 3. **Posterior collapse and its resolution.** Left: a KL warm-up ($\beta = 0$) drives the posterior to a degenerate high-KL solution that collapses the instant β is ramped in. Right: with the full recipe (free bits, $\beta = 10^{-5}$, high-noise timestep sampling) the raw KL stabilize near 40 nat and the encoder no longer collapses.

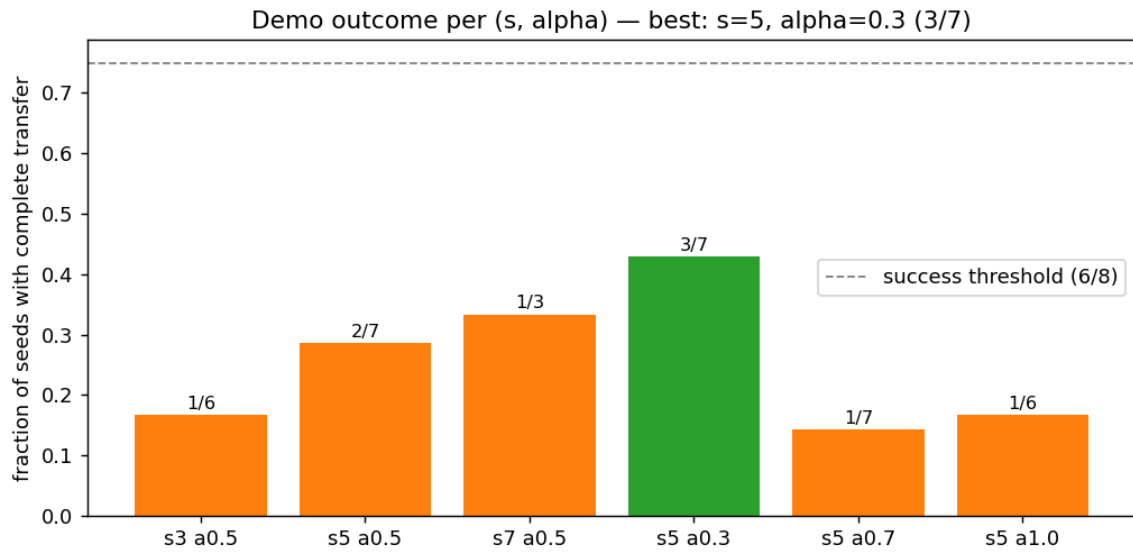


Figure 4. **Transfer outcome per (s, α)** : fraction of seeds with a complete transfer (pitch moved to B and timbre kept A on the generated output). Best at $s=5, \alpha=0.3$; higher α contaminates timbre.